CV

Dr. Danny Fainozzi

Current position: Advanced Postdoctoral Researcher, Institute of Applied Physics, University of Bern

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Education

Degree	Organisation	Duration
PhD / Dr.: Physics Dr. Claudio Masciovecchio	University of Trieste, IT Elettra Sincrotrone Trieste; thesis in collaboration with the Paul Scherrer Institute, CH	11.2017 – 10.2020 3 years
Master: Condensed Matter Physics Prof. Fulvio Parmigiani	University of Trieste, IT Thesis in collaboration with the Kyoto Institute of Technology, JP	10.2015 – 07.2017 1 year 10 months
Bachelor: Physics Prof. Fulvio Parmigiani	University of Trieste, IT Thesis in collaboration with Elettra Sincrotrone Trieste	10.2012 – 07.2015 2 years 10 months

Employment

Role	Organisation	Duration
Advanced Postdoctoral Researcher Prof. Andrea Cannizzo	University of Bern, CH Institute of Applied Physics	03.2024 – present
Junior Postdoctoral Researcher Dr. Filippo Bencivenga	Elettra Sincrotrone Trieste, IT FERMI Free-Electron Laser, TIMER endstation	04.2021 – 03.2024 3 years
Doctoral Researcher Dr. Claudio Masciovecchio	Elettra Sincrotrone Trieste, IT FERMI Free-Electron Laser (PhD, University of Trieste)	11.2017 – 10.2020 3 years

Research stays abroad

Role	Organisation	Duration
Visiting researcher Prof. Keith A. Nelson	Massachusetts Institute of Technology, US Optical Faraday-rotation measurements of nanoscale spin waves	2 months

Role	Organisation	Duration
Master thesis research Prof. Fulvio Parmigiani	Kyoto Institute of Technology, JP Master thesis project	2017

Large-scale facilities

More than ten years of experiments with femtosecond EUV and X-ray free-electron lasers, plus ultrafast optical tabletop setups.

Facility	Endstation	Role
FERMI FEL (Elettra, IT)	TIMER endstation	EUV transient gratings and EUV imaging: proposer, setup preparation, experiments, data analysis
SwissFEL (PSI, CH)		First hard X-ray transient grating (X-ray pump / optical probe): preparation, experiment, data analysis
European XFEL (DE)	FXE, MID	Long-term proposal for X-ray transient gratings: shift leader and lead data analyst
SACLA (JP)		All hard X-ray transient gratings: experiment and data analysis; campaign coordination
LCLS (SLAC, US)		All-X-ray transient grating campaign: coordination

Major achievements

Achievement 1 · Nanoscale acoustics and heat with EUV transient gratings

In a transient grating (TG) experiment two time-coincident pump pulses of wavelength λ cross on the sample at an angle 2θ and write a spatial modulation of excitation with period $\Lambda = \lambda/(2 \sin \theta)$; a delayed probe pulse diffracts from it and reads out how it evolves. Free-electron lasers extend this approach into the extreme ultraviolet [17], giving contactless access to thermoelastic dynamics at length scales no optical laser can reach.

At the TIMER endstation of the FERMI FEL I acted as proposer, setup preparator, experimentalist and data analyst, and wrote the analysis codes shared with collaborators and students. I demonstrated stimulated Brillouin scattering in the time domain at a 1 nm^{-1} wave vector, an equivalent TG period of 6 nm, the smallest pitch recorded to date [14], and tracked nanoscale heat propagation through the thermoelastic response of a SiO_2 membrane [12]. The same platform gave nanoscale polarization gratings [6], time-domain EUV diffuse-scattering spectroscopy of surface phonons [4], and the electron-lattice coupling and bilayer thermoelasticity of metal films [1, 2].

Achievement 2 · Pioneering hard X-ray transient gratings

I contributed to the first hard X-ray transient grating, an X-ray pump / optical probe proof of principle on bismuth germanate at SwissFEL [19], supported by heterodyne-detected optical TG data [15], with responsibility for preparation, execution and data analysis. I then wrote the experimental protocols for X-ray TG at FELs[8], measured X-ray circular dichroism by cross-polarized X-ray TG [7], and contributed to the theory of soft X-ray gratings [11].

The all hard X-ray transient grating (X-ray pump / X-ray probe) was demonstrated at SACLA [5], where I prepared and ran the experiment and analysed the data; this milestone was a pivot for the COST Action CA22148. Hard X-ray gratings have since launched coherent spin waves [3]. The technique is developed under a long-term proposal at

the European XFEL, where I am shift leader and lead data analyst at FXE and MID, and I coordinated campaigns at SACLA and LCLS.

Achievement 3 · Single-shot three-dimensional EUV imaging

I demonstrated stereoscopic three-dimensional coherent diffraction imaging with a single EUV FEL pulse, reaching sub-micrometre resolution [16], as experimentalist and data analyst. Single-shot 3D imaging opens a route to nanostructures in radiation-sensitive samples such as molecules, bacteria and viruses, without averaging over many identical copies.

Achievement 4 · Spin waves, phonons and EUV optics

In collaborative EUV TG campaigns I contributed to the excitation and detection of coherent nanoscale spin waves, supported by optical Faraday-rotation data I took during a research stay in Keith Nelson's group at MIT [9]; to understanding how echo interference controls phonon attenuation in nanophononic membranes [13]; to a high-efficiency programmable modulator for EUV light [10]; and to element- and enantiomer-selective real-time visualization of molecular motion [18].

Collaborations, leadership and service

- Member of COST Action CA22148 (NEXT, nonlinear EUV to hard X-ray techniques) since its start; the all-X-ray transient-grating demonstration was a pivot for its creation.
- Shift leader and lead data analyst in the European XFEL long-term proposal developing X-ray transient gratings at FXE and MID.
- Coordinator of experimental campaigns at SACLA and LCLS to advance all-X-ray transient gratings.
- Author of custom analysis codes for FEL four-wave-mixing data, shared with collaborators and students and open-sourced whenever possible.

Publications

19 selected peer-reviewed articles and preprints; first-author works marked ★. Full list on ORCID.

1. Kurdi, G., **Fainozzi, D.**, Foglia, L., Khatu, N., Laterza, S., Mincigrucci, R., … Masciovecchio, C., & Bencivenga, F. (2026). Ultrafast electron-lattice coupling in cobalt films probed by nanoscale extreme ultraviolet transient gratings. *Physical Review B 114*. doi:10.1103/j1rk-j6rt
2. Krstulović, M., Pelli-Cresi, J. S., **Fainozzi, D.**, Foglia, L., Griesi, A., … Wehinger, B., & Bencivenga, F. (2026). Thermoelastic dynamics of a nanoscale metallic film: role of bilayer coupling revealed by extreme ultraviolet transient gratings. *Journal of Applied Physics 140, 065102*. doi:10.1063/5.0340562
3. Miedaner, P. R., Maznev, A. A., Biednov, M., … **Fainozzi, D.**, … Cannizzo, A., Milne, C., Nelson, K. A., & Svetina, C. (2026). Spin waves excited by hard X-ray transient gratings. *Physical Review Letters*. doi:10.1103/zybx-dqfc
4. Capotondi, F., Maznev, A. A., Bencivenga, F., Bonetti, S., Engel, D., **Fainozzi, D.**, … & Trigo, M. (2025). Time-domain extreme-ultraviolet diffuse scattering spectroscopy of nanoscale surface phonons. *Physical Review Letters 135, 266101*. doi:10.1103/dq8w-62bm
5. Ferrari, E., Ueda, H., **Fainozzi, D.**, Osaka, T., Bencivenga, F., … & Svetina, C. (2025). All hard X-ray transient grating spectroscopy. *Communications Physics 8*. doi:10.1038/s42005-025-02178-y
6. Foglia, L., Wehinger, B., Perosa, G., Mincigrucci, R., … **Fainozzi, D.**, … Masciovecchio, C., Bonetti, S., & Bencivenga, F. (2024). Nanoscale polarization transient gratings. *Nature Communications 15*. doi:10.1038/s41467-024-54799-6

7. Rouxel, J. R., Mincigrucci, R., **Fainozzi, D.**, & Masciovecchio, C. (2024). X-ray circular dichroism measured by cross-polarization X-ray transient grating. *Journal of Physics B: Atomic, Molecular and Optical Physics*. doi:10.1088/1361-6455/ad6b62
8. **Fainozzi, D.**, Cucini, R., Vila-Comamala, J., Lima, F., … & Svetina, C. (2024). Protocols for X-ray transient grating pump/optical probe experiments at X-ray free electron lasers. *Journal of Physics B: Atomic, Molecular and Optical Physics*. doi:10.1088/1361-6455/ad717f ★
9. Miedaner, P. R., Berndt, N., Deschamps, J., **Fainozzi, D.**, … & Maznev, A. A. (2024). Excitation and detection of coherent nanoscale spin waves via extreme ultraviolet transient gratings. *Science Advances 10(36), eadp6015*. doi:10.1126/sciadv.adp6015
10. Vaskivskyi, I., Mraz, A., Venturini, R., … **Fainozzi, D.**, Bencivenga, F., Masciovecchio, C., & Mihailovic, D. (2024). A high-efficiency programmable modulator for extreme ultraviolet light with nanometre feature size based on an electronic phase transition. *Nature Photonics*. doi:10.1038/s41566-024-01389-z
11. Foglia, L., Mincigrucci, R., Döring, F., **Fainozzi, D.**, Mamyrbayev, T., Masciovecchio, C., & Bencivenga, F. (2024). A new framework for soft X-ray transient gratings. *Journal of Physics B: Atomic, Molecular and Optical Physics*. doi:10.1088/1361-6455/ad4fd0
12. **Fainozzi, D.**, Caporaletti, F., Capotondi, F., De Angelis, D., … Bencivenga, F., & Baldi, G. (2024). Temperature effects on the nanoscale thermoelastic response of a SiO_2 membrane. *APL Materials 12(5)*. doi:10.1063/5.0201769 ★
13. Hadi, M., Luo, H., Pailhès, S., Tanguy, A., … **Fainozzi, D.**, … & Giordano, V. M. (2024). The effect of echoes interference on phonon attenuation in a nanophononic membrane. *Nature Communications 15, 1317*. doi:10.1038/s41467-024-45571-x
14. **Fainozzi, D.**, Foglia, L., Khatu, N. N., Masciovecchio, C., Mincigrucci, R., Paltanin, E., & Bencivenga, F. (2024). Stimulated Brillouin scattering in the time domain at 1 nm^{-1} wave vector. *Physical Review Letters 132, 033802*. doi:10.1103/PhysRevLett.132.033802 ★
15. **Fainozzi, D.**, Catalini, S., Torre, R., Masciovecchio, C., & Svetina, C. (2023). BGO relaxation dynamics probed with heterodyne detected optical transient gratings. *arXiv:2309.02308*. Preprint. doi:10.48550/arXiv.2309.02308 ★
16. **Fainozzi, D.**, Ippoliti, M., Bille, F., De Angelis, D., … Bencivenga, F., & Capotondi, F. (2023). Three-dimensional coherent diffraction snapshot imaging using extreme-ultraviolet radiation from a free electron laser. *Optica 10(8), 1053–1058*. doi:10.1364/OPTICA.492730 ★
17. Foglia, L., Mincigrucci, R., Maznev, A. A., Baldi, G., … **Fainozzi, D.**, … & Bencivenga, F. (2023). Extreme ultraviolet transient gratings: a tool for nanoscale photoacoustics. *Photoacoustics 29, 100453*. doi:10.1016/j.pacs.2023.100453
18. Mincigrucci, R., Rouxel, J. R., Rossi, B., … **Fainozzi, D.**, … Chergui, M., & Masciovecchio, C. (2023). Element- and enantiomer-selective visualization of molecular motion in real-time. *Nature Communications 14, 386*. doi:10.1038/s41467-023-36047-5
19. Rouxel, J. R., **Fainozzi, D.**, Mankowsky, R., Rösner, B., … & Svetina, C. (2021). Hard X-ray transient grating spectroscopy on bismuth germanate. *Nature Photonics 15, 499–503*. doi:10.1038/s41566-021-00797-9